FIT5202 S1 2026 Assignment 1: Analysing Road Safety Data (Weight: 15%)

Feedback Sheet

Student Name : Bharath Gandhimani (35501308)						Comments	Marks Awarded	Total Sub Section
Marked By : Arthur								
			N(0%)	P(50%)	F(100%)			
Part 1	**Data Loading and Transformation (15%)**							
	Section Breakdown							
1. Data Preparation and Loading	1.1	Write the code to create a SparkContext object using SparkSession. Use the UK (Europe/London) time as the session time zone. Give your application an appropriate name and run Spark locally with 4 cores on your machine.	☐	☐	☑	Good	1	
	1.2	Load the CSV and JSON files into multiple data frames with an explicit StructType schema (most of the columns can use an integer, except for those with text data and latitude/longitude).	☐	☐	☑		2	
	1.3	Display the total row count and the first 5 records of the collision, accident and vehicle.	☐	☐	☑		2	

2. Data cleansing	2.1	Handle Missing/Invalid Values: Filter the DataFrames to remove any records where key IDs are null and where did_police_officer_attend_scene_of_accident contains the missing data code -1.	☐	☐	☑		2	15
	2.2	Create the specified UDF for spatial bounding box.	☐	☐	☑		2	
	2.3a	Check and print t	☐	☐	☑		2	
	2.3b	Explain how PySpark determines the default number of partitions when reading a CSV file from a local file system.	☐	☐	☑		2	
	2.3c	If we plan to run heavy aggregations grouping by year, month and weather conditions, what strategies should we use to partition our data?	☐	☐	☑		2	
Part 2	**Working with DataFrames (50%)**							
	Section Breakdown							

	1.1	Using native PySpark functions, convert the date column into a proper PySpark DateType. Then, extract the Year and Month into two new integer columns: year and month. Show the first 5 rows of these specific columns.	☐	☐	☑	Good	5	
	1.2	Perform a join between the collision data frame and the police_force ID mapping, and add the string name of the police force to your collision data frame.	☐	☐	☑		5	

	1.3	Filter the collision data frame to isolate incidents that occurred during "darkness" (using light_conditions) AND while it was "raining or snowing" (weather_conditions). Calculate the ratio of severe/fatal collisions to slight collisions under these exact conditions and compare it to the same ratio during clear daylight.	☐	☐	☑	The appropriate method for reading metadata was not used: manually hardcoded	4	
1.Query and Analysis	1.4	Use the vehicle data frame to find young drivers (aged 17-21) driving small-engine cars (engine_capacity_cc <= 1600). What are the top 3 most common vehicle_manoeuvre types for this demographic that resulted in the vehicle leaving the carriageway (vehicle_leaving_carriageway not equal to 0)?	☐	☐	☑	The appropriate method for reading metadata was not used:	4	45

1.5	Join the accidents (casualties) table with the vehicle table to identify which specific vehicle_type (by string name) is statistically most likely to be involved when a pedestrian (casualty_class = 3) is fatally injured.	☐	☐	☑	The appropriate method for reading metadata was not used:	4
1.6	Join tables and perform aggregations: add two columns to the collision dataframe: number_of_vehicles, number_of_casualties.	☐	☐	☑		5
1.7	Group the data to find the total number of collisions for each police_force and road_type. Print the highest road_type for each police_force.	☐	☐	☑	The appropriate method for reading metadata was not used:	4
1.8	Group collisions by local_authority_district, and create a custom "Danger Index".	☐	☐	☑	The appropriate method for reading metadata was not used:	4

	1.9	(Open Question) Explore the dataset freely and plot two diagrams of your choice. Which attributes are highly correlated with the number and severity of accident casualties?	☐	☐	☑		10	
Part 3	**Query Optimisation (35%)**							
	Section Breakdown:							
1. Query Implementation	3.1a	Implement the complex query in DataFrame API	☐	☑	☐	Incorrect Vechicle_type included	2.5	30
	3.1b	Implement the complex query in SQL	☐	☑	☐		2.5	
	3.2	Explore the DAG in Spark UI and discuss the bottle neck	☐	☐	☑		10	
2. Query optimisation	3.3	Implement your suggested improvements in 3.2, run the query, measure the time using the built-in magic command and showcase the improvements.	☐	☐	☑		15	
Additional: Quality Aspect								
Code Quality	Follows programming standards, readability of the code, organization of code. Please find the PEP 8 -- Style Guide for Python Code for your reference. Here is the link: https://peps.python.org/pep-0008/		☑	☐	☐		0	0

	The output of the code should be displayed along with the code when submitting the jupyter notebook file	☑	☐	☐		0	
(note: Code Quality is a penalty section, 0 means you did well and didn't lose mark.)					**Subtotal:**		**90**
					Late Submission Day(s)	0	0.00
Final Grade					**HD**		90.00
Contribution to Final Marks							**13.5**